40 — Findings Register, Risks & Task List

v1.4 — the honest state of the program, both tracks

Function: The consolidated register of every finding that materially changed the numbers or the architecture — solid-track F1–F5 and cross-track X1–X12 — plus the correction trail, the task list, and the make-or-break bench questions. **Anyone reconstructing this concept should read this document before trusting any sizing figure.** The physics backbone holds on multiple independent passes; quantitative closure is bench-gated exactly where marked.

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1. Solid-track findings (F-series)

F1 — M-cycle working-air moisture is the dominant latent term

The sink (29 °C) cannot absorb heat from a 25 °C cabin, so all cabin sensible load exits evaporatively via the M-cycle working air; that moisture must pass through the desiccant (exhaust-and-replace in open cycle, recycle in X8 mode — equivalent within ~10–16% at DP-A). **Effect: peak duty ~9–11 kg/h, not ~2 kg/h; regen ~7.5–9 kW continuous; condenser/fan duty scale with it.** The self-consistent steady state (doc 00 §4) sharpened the original 7–8 kg/h estimate — the hand-chained cabin state was not self-consistent. Closure: the moisture returns as condensate covering M-cycle feed with ~50–70 L/day surplus; the system is a coherent heat-driven chiller. Full transient T2 sets the final charge.

F2 — Solar-direct regeneration is equilibrium-dead at the peak point

Against a condensing purge (~25 g/kg), a 45–50 °C bed faces ~29–38% RH at its face — at/above AlFu's step: zero driving force; no kinetic result rescues it (table, doc 20 §6). Heat-source decision: waste-heat primary; the **liquid track is the solar comfort island** (X2), with heat-pump lift retained behind it. M2 *confirms* F2; its real deliverables are 60–65 °C completeness/kinetics and effective Δq.

F3 — The sizing Δq is optimistic

0.2–0.3 g/g is a **dry-purge full-swing** figure; a plausible 0.05–0.10 g/g residual at 60–65 °C against humid purge cuts effective Δq to ~0.15–0.2 g/g (+25–50% inventory), compounding F1. All sizing tables carry the annotation; M2 extracts effective Δq as a first-class output.

F4 — Mill-zero needs a feedstock-reactivity branch

The literature mortar-and-pestle demonstration used *freshly precipitated* amorphous $Al(OH)_3$; crystalline tech-grade gibbsite is kinetically sluggish. A PXRD fail may be feedstock, not milling energy. Mitigations, cheapest first: sieve fine → extend aging toward 120 °C → **dissolve-and-reprecipitate fresh gel** (an evening vs a mill build). Integrated into the doc 21 §4 tree.

F5 — Aluminum DCHX in chloride service

The coated aluminum exchanger sits in salt-bearing intake air with brine and raw water elsewhere on the platform — the materials class the two-worlds rule prohibits near chloride. Mitigations decided on paper (doc 22 §3): sealed/filtered intake, no dissimilar-metal fittings, coating treated as a barrier layer with edge/defect behavior in M4's scope, materials law imported from the liquid track — and **X8 closure as the primary mitigation** (the working path ingests no ambient aerosol). Corrosion *rate* is M4's.

Minor notes

- **PVA anneal:** the 120–150 °C activation doubles as the mandatory anneal (doc 22 §2) — a re-coat never skips it.
- **Ten-minute cycling** thermally cycles the fluid inventory and manifolds; the sensible bucket may be optimistic for the plumbing (doc 20 §4; M3 blank measures it).

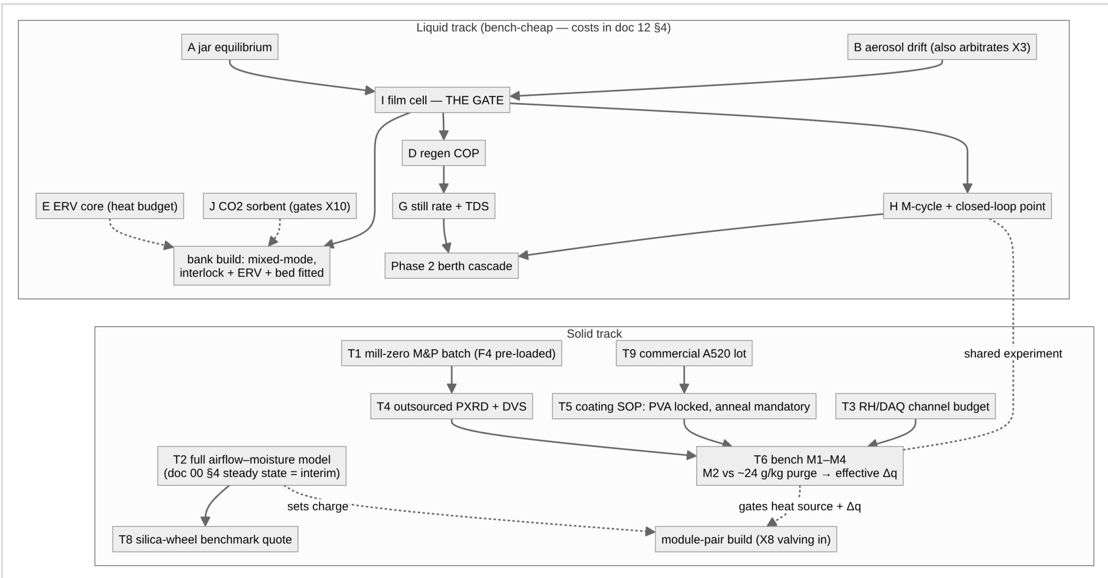
2. Cross-track findings (X-series)

#	Finding (one line)	Where it lives now
X1	Once-through ventilation and whole-cabin M-cycle cooling never compose (5.5–11 kg/h absorber duty at DP-A); whole-cabin liquid AC deferred — in recirc topology it converges to solid-track-class duty (~10.6 kg/h, ~11 kW)	doc 00 §4/§6; doc 10 §2
X2	Regeneration asymmetry is a theorem: the sealed still (no purge) keeps positive driving force at any pool >~40 °C; the solid bed hits F2's wall below ~50 °C → **liquid = solar layer, solid = waste-heat layer**	doc 00 §3; doc 30 §2
X3	The old blanket rejection of liquid desiccants for cabin air is re-scoped: "rejected **absent demonstrated aerosol control** — test B is the arbiter"	doc 21 §2; doc 11 §3
X4	Materials doctrine flows both ways: nylon ban, brazed-plate prohibition, drip discipline, sourcing heuristic imported to the solid track	doc 22 §3
X5	The airflow cascade: route makeup through the cabin before it becomes working air — ventilation becomes latently free. Berth scale: full 48 m³/h kept (~1,920 vs ~2,670 ppm), *more* cooling (102 vs 86 W), 1.6 °C supply cost. Solid open cycle: ~600 ppm automatic at F1-scale flows	doc 10 §2; doc 20 §2
X6	Mixed-mode recirculation (CO_2-interlocked fresh minimum + recirculated bulk) is the liquid occupied baseline — once-through fails 4-adult comfort at DP-A (59–66% RH); mixed-mode restores 55% at 0.88 kg/h. Recirc-only prohibited	doc 10 §2; doc 00 §5

#	Finding (one line)	Where it lives now
X7	ERV latent recovery on the fresh stream: (1–ε)×10.4 g/kg per kg; ε ≥0.8 saves ~9–10 kWh/day at DP-A and makes generous ventilation cheap. Requires one ducted exhaust path; CO_2 crossover <5% (test E). Applies to both tracks	doc 11 §5; doc 20 §2
X8	Exhaust-vapor recovery doctrine: saturated process exhausts terminate in distillation recovery; no stream is both rejection sink and recovery source (protects the ERV); closed working-air loop = solid design intent (water-neutral in every ambient) and liquid free-heat option (~1.3 kWh/L); air-swept regen gains a condenser (technical water PENDING TDS)	doc 00 §7; doc 20 §8; doc 11 §4
X9	All-electric galley, no gas in the envelope: one burner = 3.6× the crew's CO_2 + ~0.25 kg/h latent. ~2 kWh_e/day induction	doc 00 §5; doc 30 §4
X10	The CO_2 battery: two-bed solid-amine TSA, 85–95 °C regen, ~1.6 kg/day for ~1.6–2.5 kWh/day — the only path holding <1,000 ppm sealed.**Required-PENDING test J** (slip assay gates cabin connection); fallback = oversized ERV+DCV, open-cycle only	doc 00 §5; doc 11 §5; doc 12 §4
X11	Sealed-mode CO_2-bed chemistry is heat-grade-dependent: K_2CO_3/apolar-carbon TSA (regen ~130–150 °C; alumina/MgO supports prohibited — double-salt deactivation; breathing-air gate = alkaline-carryover assay, **required-PENDING test J-K**) is the waste-heat-platform primary; the solid-amine resin (85–95 °C) is the solar-grade bed. The ladder runs ventilation-first from the top of heat availability down; a small amine bed is retained as outage fallback where potash is primary (the ETC's ≤93 °C cannot regenerate carbonate)	doc 00 §5; doc 12 §4–6; doc 30 §2/§6
X12	The brine's activity depression is a **temperature lift**, not only a humidity floor (the dual of X2): a single-stage $CaCl_2$ absorption heat transformer — with the sealed still unchanged as its generator + condenser — upgrades a 60–65 °C waste tail to 85–90 °C at COP ~0.45–0.48 with no electricity beyond transfer pumps, reaching amine-bed and hot-regen grade. Lift ceiling aw(x)·Psat(T_abs) < Psat(T_evap) → ≤91 °C single-stage (potash grade out of reach). **Upgrade path only, never load-bearing; PENDING test L**	doc 31 §2; doc 12 §4

Specification P17 (safety-critical): CO_2 <1,000 ppm at all times, all modes; alarm + forced boost 2,000 ppm; per-room max sensing; mechanical minimum stop. Full register: doc 00 §8.

3. Dependency structure and task list



Ranking: liquid A/B/E/J and solid T1/T9 all run in parallel at trivial cost; test I is the liquid gate; T2 gates every solid downstream size. Shared: test H and the solid M-cycle validation are one experiment; the wet/dry-bulb instrumentation stack (doc 22 §7) serves both. **T7 (documentation pass) is closed by this lineage.** Deferred: CAU-23/CAU-10-H, the CO_2-sorbent watch register (doc 12 §6 — MOF-808-AA primary; tetraamine-Mg_2(dobpdc), superseding the diamine class), two-core sensible recovery, the hybrid brine-pre-stage-before-AlFu architecture (worth revisiting only with hot-regen brine — the polish split favors it only then). On waste-heat platform variants, test J-K joins the parallel cheap set (X11), sharing J's rig and instrumentation. The **doc 31 upgrade paths** (X12 AHT first; coupled VC heat pump; still MVR; closed AlFu chiller; static crystallizer pot) are boost modes on the deferred list — additions riding on the passive baseline, never core; tests **L** and **A3** (doc 12 §4) join the cheap parallel set only when their platform trigger exists (waste-heat tail / mass-limited reserve respectively), and the prime-mover heat-pump variant is recorded rejected (doc 31 §1).

4. Make-or-break bench questions (consolidated)

- ☐ Real $CaCl_2$ aw vs wt% and T — 40 wt% @ 33 °C ≤ 55% ERH (A)
- ☐ Aerosol chain: clean steel coupon, trickle and flooded (B — gates every cabin connection and arbitrates X3)
- ☐ Film K·a, wetting threshold, staged NTU/m, mixed-mode flow point (I)
- ☐ ERV real ε_lat, salt fouling, CO_2 crossover <5% (E)
- ☐ CO_2 sorbent: capacity at 1,000–1,500 ppm / 45–55% RH, 85–95 °C completeness, water co-adsorption, **amine/ammonia slip**, cycle fade (J)
- ☐ Potash arm (X11): K_2CO_3/carbon capacity at 1,000–1,500 ppm / 45–55% RH; 120/135/150 °C completeness with logged purge; alumina-deactivation control; caking/deliquescence; **alkaline carryover**; cycle stability (J-K)
- ☐ Effective working Δq under ~24 g/kg purge at 60–65 °C (F3 · M2)
- ☐ Desorption completeness/rate in the 10-min half-cycle (F2 · M2)

- ☐ Specific regeneration energy vs 0.8–1.0 kWh/L, incl. plumbing overhead (M3)
- ☐ Coating adhesion, anneal, capacity retention, **coated-face ΔP** (M1, T5)
- ☐ Coating durability + salt-air edge behavior over 10^2–10^3 swings (M4 · F5)
- ☐ Still rate + condensate TDS ≈ 0; air-swept-condenser TDS (G)
- ☐ M-cycle wetting/heel + closed-loop feed point (H — shared)
- ☐ Distillate potability lab test before regular drinking
- ☐ T2 outputs: charge, coated area, condenser duty, fan power at the self-consistent peak; correction path (cycling vs area vs setpoint)
- ☐ **Coated-area denominator** — doc 22 §4's sizing table implies ~0.3 kg/m² where §2 specifies 0.18 kg/m² planform (one-side vs two-side convention unstated). Settled empirically by M1 on a known geometry; size from 0.18 until then (doc 22 §4)
- ☐ **CO_2-battery duty basis** — the ~1.6 kg/day figure is a 16 h/8 h daily average; the all-awake peak at the 48 m³/h floor is ~2.05 kg/day, and the published 91 m³/h at 50% capture independently implies ~2.16 kg/day. P17 is an at-all-times spec, so bed mass and regen heat want the peak (test J)
- ☐ **One envelope-leakage assumption for both tracks** — liquid sizes on 280 g/h (occupants only), solid on 1.8–2.8 kg/h (envelope + occupant), implying ~0.95 ACH where doc 12 §3 uses 0.05–0.15 for the unattended case. Bounded: duty moves 8.8→10.3 kg/h across that range, so F1 holds either way
- ☐ **Contactor rating stated with its approach and flow** — the 0.4–0.8 kg/h rating reproduces at the ~0.19 m/s DP-A baseline, not at the stated 0.6–1.0 m/s band; the NTU-1.9/85% approach used for depth does not hold at the higher velocity (test I)
- ☐ **Real ε_lat behind the ERV'd duty line** — doc 10 §3 and doc 12 §1 carry 17.4 and 15.9 g/kg for the same pre-dried state, i.e. ε ≈ 0.65 vs the specified ≥0.8; the published ERV'd removal tracks 17.4. Settled by test E (doc 12 §2 erratum 10). The bare-fresh 0.88 kg/h peak governs sizing either way

5. Correction-trail principle

Every erratum stays visible (doc 12 §2 carries the liquid trail; F1's sharpening from 7–8 to 9–11 kg/h is recorded above). The correction trail is part of the design's credibility: *never cite a margin as a computed value; re-run every comfort claim when the design point moves; solve steady states simultaneously, never as hand chains.*

The parameter register (doc 50 §3.5) is the standing mechanism for this: it re-derives every headline figure from its stated basis, so a number that has drifted from the basis printed beside it surfaces on its `Checks` sheet rather than surviving unexamined. Reconciling that sheet is a release gate (doc 50 §8). Its first pass produced the two open items above, erratum 10, and a set of basis-labelling clarifications across docs 00/10/12/20/22 — no design figure moved. *Added lesson: a quantity is only as good as the basis printed next to it; state the basis, or the number will eventually be read on the wrong one.*

Version history

- **v1.0** — New lineage: archived 05 (F1–F4) merged with the archived doc 06 X-register (X1–X10, F5, P-series patches now applied in place across docs 00–30); task T7 closed by this lineage; dependency chart unified across tracks.
- **v1.1** — X11 recorded; test J-K added to the make-or-break list and task notes; deferred list updated to the doc 12 §6 watch register.
- **v1.2** — X12 ($CaCl_2$ absorption heat transformer) recorded; doc 31 upgrade-path family added to the deferred notes with tests L/A3 pointers and the prime-mover heat-pump rejection; X-range extended in the function statement.
- **v1.4** — Dependency-chart label follows doc 12 v1.4 in dropping the retired aggregate cost headline; four verification findings added to §4 (CO_2-battery duty basis, the X2 crossover, the cross-track latent-gains basis, and the contactor rating/velocity/approach triangle).
- **v1.3** — First parameter-register reconciliation pass (doc 50 §3.5): two open items added to the make-or-break list — the doc 22 coated-area denominator (gated on M1) and the ε_lat inconsistency behind the ERV'd duty line (gated on test E, doc 12 §2 erratum 10) — and §5 records the register as the standing mechanism for catching basis drift. No finding ID renumbered; no design figure changed.